

In the formulation of statistical mechanics, Gibbs introduced three standard ensembles to which real experiments could be approximated. These are: (i) the microcanonical ensemble, (ii) the canonical ensemble and (iii) the grand canonical ensemble. Their classification depends on the manner in which their systems interact. Each of them has its own characteristics distribution. Physical systems can interact in a variety of ways; in particular, they may exchange energy, or matter or both with each other. The ensemble in which the systems exchange energy but not matter, is called a *canonical ensemble*, that in which both energy and matter are exchanged between the systems is called a *grand canonical ensemble*; and that in which neither energy nor matter is exchanged, is called a *microcanonical ensemble*. Statistical mechanics provides a number of methods for calculating equilibrium thermodynamic properties of macroscopic systems. Explicit calculations of thermodynamics functions can be carried out using microcanonical, canonical or grand canonical ensembles. A specific choice of ensembles may be thought of as corresponding to a particular physical situation.

5.1 MICROCANONICAL ENSEMBLE

This ensemble consists of systems which are isolated from the rest of the world. Such a system is also known as a 'closed isolated system' and has a fixed volume, fixed total energy and a fixed total number of particles. The probability density $P(p, q)$ of such a system differs from zero only on the constant energy hypersurface. It must be noted, however, that in reality one cannot achieve complete isolation of a system. We must allow for some interaction energy ΔE , though very small. The elements of the microcanonical ensemble, therefore, lie within the range between E and $E + \Delta E$.

The picture of a microcanonical distribution in phase space would be something like a very thin uniform cloud. One can easily see this by referring to Fig. 5.1, which gives the trajectory of a simple harmonic oscillator. The ellipse E represents the evolution of our conservative system with total energy E . This system is just one member of the ensemble. The other members may have energy between E and, $E + \delta E$

and hence their corresponding phase points will be between two ellipses. The thin elliptic cloud is the energy shell. If we conceive of δE as getting smaller and smaller, the energy shell between the two surfaces would become just a surface in the limit as $\delta E \rightarrow 0$.

According to the fundamental postulate of equal a priori probability under the condition of equilibrium the system is equally likely to be found in one of its accessible states. In the case of microcanonical ensemble all states between E and $E + \delta E$ are equally accessible. Therefore, if the

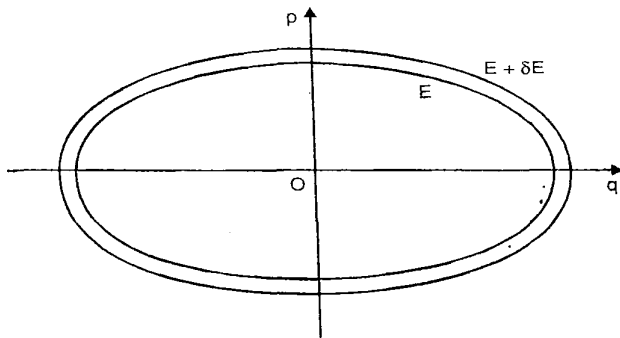


Fig. 5.1

system is in a state X , corresponding to the energy E_x , the probability P_x of finding the system in state X is given by

$$P_x = \begin{cases} C & \text{if } E < E_x < E + \delta E \\ 0 & \text{otherwise} \end{cases} \quad \dots(5.1)$$

where C is a constant, the value of which can be determined from the normalization condition $\sum P_x = 1$, when summed over all accessible states in the range between E and $E + \delta E$. It should be more appropriate to write this relation as $P_x = C \delta(E - E_x)$, where δ is the Dirac delta function.

5.2 CANONICAL ENSEMBLE

It is evident that the microcanonical ensemble is fundamental and refers to the simple situation in which the constituent system are isolated and not influenced in any way by external disturbances. One would, therefore, think it proper to use such ensembles in the development of the basic theory. However, as pointed out above, microcanonical ensembles are best suited for isolated systems, and we seldom deal with a completely isolated system; there is always some energy exchange with surroundings. It would be more realistic, therefore, to use an ensemble in which statistical equilibrium is attained not by isolation but by free interaction. Canonical ensembles have been found to be more appropriate

for the description of such systems. Besides, canonical approach gives results which apply even when components interact strongly.

Suppose we desire to study the statistical properties of a thermodynamical system A . To this end we envisage an assembly of a very large number of systems (N) identical in volume and molecular composition with A . Each system of the assembly is considered to be in thermal contact with a huge heat reservoir. A heat reservoir is a system whose heat capacity is so much higher than that of the subsystem in contact with it, that heat flow from or to the heat reservoir does not change its temperature significantly. The assembly can be thought of as contained within a microcanonical ensemble. Each system of the assembly is in contact with the other elements of the microcanonical ensemble, but is otherwise isolated from the outside world. The walls of the containers of the system of the assembly which separate it from microcanonical ensemble are *diathermic* walls, which allow heat to flow through them but do not allow the exchange of mass. An assembly of this kind is called a *canonical ensemble* of systems of type A . The number of systems in a canonical ensemble is constant; but their energy is not fixed.

Let us now find the probability that a canonical system, under the condition of equilibrium, is in a particular state with specific energy E_α .

Consider a canonical system A in a microcanonical ensemble A' ($A \ll A'$) (Fig., 5.2). The walls of A are such that A and A' are free to exchange energy. The microcanonical ensemble is isolated and its total energy E_0 is constant. As discussed before, a large number of microscopic states correspond to the macroscopic state of the microcanonical ensemble with energy E_0 . Let Ω_0 be the total number of microstates of the microcanonical ensemble with energy E_0 . By the postulate of equal a priori probability, all these microstates are

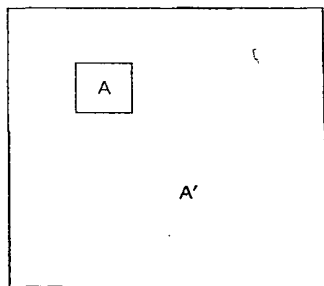


Fig. 5.2

equally probable. Therefore, the probability of each microstate is $\frac{1}{\Omega_0}$. It

is obvious that different microstates of the microcanonical ensemble A' will give rise to different microstates of the small subsystem A . Suppose, a certain microstate of A' gives rise to a microstates of A with energy E_α ($E_\alpha \ll E_0$). Hence, the energy of the remaining part of the complete system is $E_0 - E_\alpha$. This is one microstate of A with energy E_α , attained through the other microstates of A' . Let the number of states of the microcanonical ensemble A' , through which the states of the small system A corresponding to the energy E_α are attained, be $\Omega_\alpha (E_0 - E_\alpha)$. Therefore, the probability that the small system A is in a state with energy E_α is

$$P_{\alpha} = \frac{\Omega_{\alpha}(E_0 - E_{\alpha})}{\Omega_0} \quad \dots(5.2)$$

Using the relation

$$a = e^{\ln a}$$

we can express Eq. (5.2) as

$$P_{\alpha} = \frac{e^{\ln \Omega_{\alpha}(E_0 - E_{\alpha})}}{\Omega_0} \quad \dots(5.3)$$

Expanding $\ln \Omega_{\alpha}(E_0 - E_{\alpha})$ into a Taylor's series

$$\ln \Omega_{\alpha}(E_0 - E_{\alpha}) = \ln \Omega_{\alpha}(E_0) - E_{\alpha} \frac{\partial \ln \Omega_{\alpha}(E_0)}{\partial E_0} + \text{higher order terms} \quad \dots(5.4)$$

Putting $\beta = \frac{\partial \ln \Omega_{\alpha}(E_0)}{\partial E_0}$ and neglecting higher order terms

$$\ln \Omega_{\alpha}(E_0 - E_{\alpha}) = \ln \Omega_{\alpha}(E_0) - \beta E_{\alpha}$$

Hence

$$\begin{aligned} P_{\alpha} &= \frac{e^{(\ln \Omega_{\alpha}(E_0) - \beta E_{\alpha})}}{\Omega_0} = \frac{e^{\ln \Omega_{\alpha}(E_0)}}{\Omega_0} e^{-\beta E_{\alpha}} \\ &= \frac{\Omega_{\alpha}(E_0)}{\Omega_0} e^{-\beta E_{\alpha}} = \text{const.} \times e^{-\beta E_{\alpha}} \quad \dots(5.5) \end{aligned}$$

From the normalization condition $\sum P_{\alpha} = \text{const.} \times \sum e^{-\beta E_{\alpha}} = 1$

$$\text{Therefore,} \quad \text{const.} = \frac{1}{\sum e^{-\beta E_{\alpha}}} \quad \dots(5.6)$$

and

$$P_{\alpha} = \frac{e^{-\beta E_{\alpha}}}{\sum e^{-\beta E_{\alpha}}} \quad \dots(5.7)$$

This is the Gibbs *canonical distribution*. It gives the probability that the small system A be in a state with energy E_{α} . There may be other microstates of A with the same energy E_{α} . If this number is g_{α} .

$$P_{\alpha} = \frac{g_{\alpha} e^{-\beta E_{\alpha}}}{\sum e^{-\beta E_{\alpha}}} \quad \dots(5.8)$$

The denominator is usually represented by Z , i.e.

$$Z = \sum_{\alpha} e^{-\beta E_{\alpha}} \quad \dots(5.9)$$

Therefore

$$P_{\alpha} = \frac{g_{\alpha} e^{-\beta E_{\alpha}}}{Z} \quad \dots(5.10)$$

If E is a continuous variable

$$Z = \int_0^{\infty} e^{-\beta E} dE_{\alpha} \quad \dots(5.11)$$

We have yet to identify the parameter β , which we will shortly do. Meanwhile, we shall discuss here an example which gives a clue towards the identification of β .

Consider a composite system A^0 made up of two systems A and A' in thermal contact and weakly interacting with each other. Let A be in a state r with energy E_r and A' in a state s with energy E_s . The energy of the composite system A^0 , therefore, is $E^0 = E_r + E_s$.

If the two systems are separately in internal equilibrium, the probabilities of finding them in respective states are given by

$$P_r = \frac{e^{-\beta E_r}}{\sum_r e^{-\beta E_r}}, \quad P_s = \frac{e^{-\beta E_s}}{\sum_s e^{-\beta E_s}} \quad \dots(5.12)$$

These probabilities are statistically independent. Therefore, the probability P_{rs} that the system A is in state r and A' in state s simultaneously, is given by the product of P_r and P_s .

$$P_{rs} = P_r P_s = \frac{e^{-\beta E_r}}{\sum_r e^{-\beta E_r}} \frac{e^{-\beta E_s}}{\sum_s e^{-\beta E_s}} \quad \dots(5.13)$$

Evidently, this does not correspond to a canonical distribution of the composite system and, hence, does not describe an equilibrium situation. Exchange of energy between the two systems, therefore will continue until equilibrium is reached. The time taken by the systems to reach the state of equilibrium from its initial state of non-equilibrium is called relaxation time.

However if $\beta = \beta^1$. Eq. (5.13) reduces to

$$P_{rs} = \frac{e^{-\beta(E_r + E_s)}}{\sum_r \sum_s e^{-\beta(E_r + E_s)}} \quad \dots(5.14)$$

which corresponds to the canonical distribution.

Since the only criterion for thermodynamic equilibrium under these conditions is that the temperature of the two systems be equal, it is valid to assume β as a function of temperature.

5.3 ALTERNATIVE METHOD FOR THE DERIVATION OF CANONICAL DISTRIBUTION

Consider an ensemble consisting of a large number of systems N . As in the preceding section the system are separated by diathermic walls permit the transfer of energy from one to the other. Suppose now that

n_1 of these systems are in state 1 with energy E_1 ,

n_2 of these systems are in state 2 with energy E_2 ,

n_3 of these system are in state 3 with energy E_3

.....

n_i of these systems are in state i with energy E_i

...(5.15)

This is one of the modes of distributing the total energy E among N constituents of the ensemble. It is obvious that

$$\sum n_i = N; \quad \sum n_i E_i = N \langle E \rangle \quad \dots(5.16)$$

where $\langle E \rangle$ stands for the average energy per system.

Any set of numbers $(n_1, n_2, \dots, n_i)$ such as (5.15) which satisfies the restrictive condition given by Eq. (5.16) is a possible mode of distribution and any such mode can be realized in a number of ways by effecting a reshuffle among the members of the ensemble.

We will now find the probability for such a distribution to occur.

Let us first find the number of ways in which a distribution such as (5.15) can be obtained. Since, of the total number of N systems n_1 are to be accommodated in state 1, the first can be chosen in N ways; the second in $N - 1$ ways ... and n_1 th $N - n_1 + 1$ ways.

Therefore, the total number of possible ways = $N(N - 1) \dots (N - n_1 + 1)$

$$= \frac{N!}{(N - n_1)!} \quad \dots(5.17)$$

However, n_1 systems in state 1 are distinguishable and, hence, permutations of these systems among themselves do not give a new mode of distribution. The number of possible permutations being $n_1!$, the total number of distinct ways in which n_1 systems can be selected out of N is

$$\frac{N!}{n_1!(N - n_1)!}$$

We are now left with $N - n_1$ systems, of which n_2 are to be placed in state 2. This can be done in $\frac{(N - 1)!}{n_2!(N - n_1 - n_2)!}$ ways.

Proceeding in this way we find that the number of ways in which the distribution such as (5.15) can be obtained is

$$\begin{aligned} &= \frac{N!}{n_1!(N - n_1)!} \times \frac{(N - 1)!}{n_2!(N - n_1 - n_2)!} \times \\ &\quad \frac{(N - n_1 - n_2)!}{n_3!(N - n_1 - n_2 - n_3)!} \times \dots \\ &= \frac{N!}{n_1! n_2! n_3!} \quad \dots(5.18) \end{aligned}$$

The probability W of obtaining such a distribution can be found from the ratio of this number to the total number of all possible distributions, say C :

$$\text{Therefore} \quad W = \frac{1}{C} \frac{N!}{n_1! n_2! \dots} \quad \dots(5.19)$$

Accordingly, the most probable mode of distribution will be the one for which W is maximum.

Taking logarithms

$$\ln W = \ln N! - \sum_i n_i \ln n_i - \ln C \quad \dots(5.20)$$

By Stirling's formula

$$\ln N! = N \ln N - N$$

$$\text{Therefore,} \quad \ln W = N \ln N - N - \sum_i n_i \ln n_i + \sum_i n_i - \ln C$$

$$\ln W = N \ln N - \sum_i n_i - \ln C \quad \dots(5.21)$$

For W to be maximum $\delta \ln W = 0$ i.e.

$$\delta \sum_i n_i \ln n_i = 0$$

$$\text{Therefore } \sum (\delta n_i + \ln n_i \delta n_i) = 0 \quad \dots(5.22)$$

The other restrictive conditions are

$$\sum_i n_i = N \text{ and } \sum_i \frac{n_i E_i}{N} = E$$

On taking differentials

$$\sum_i \delta n_i = 0; \quad \sum_i E_i \delta n_i = 0 \quad \dots(5.23)$$

Therefore, for W to be maximum

$$\sum_i \ln n_i \delta n_i = 0 \quad \dots(5.24)$$

subject to the conditions (5.23).

Following the technique of Lagrange's multipliers (see Appendix A), we multiply the first of Eq. (5.23) by α and the second by β and add these to Eq. (5.24), to get

$$\sum_i (\ln n_i + \alpha + \beta E_i) \delta n_i = 0 \quad \dots(5.25)$$

If α , β are chosen suitably, δn_i can be regarded as independent and, hence, the coefficient of each δn_i must vanish separately.

$$\text{Therefore } \ln n_i + \alpha + \beta E_i = 0 \quad \dots(5.26)$$

$$\text{or} \quad n_i = e^{-\alpha - \beta E_i} = e^{-\alpha} e^{-\beta E_i} \quad \dots(5.27)$$

Since

$$\sum_i n_i = N$$

$$e^{-\alpha} \sum_i e^{-\beta E_i} N \quad \dots(5.28)$$

and

$$n_i = \frac{N e^{-\beta E_i}}{\sum_i e^{-\beta E_i}} \quad \dots(5.29)$$

The probability of finding a system in state i is $P_i = \frac{n_i}{N}$

$$\text{Therefore} \quad P_i = \frac{n_i}{N} = \frac{e^{-\beta E_i}}{\sum_i e^{-\beta E_i}} \quad \dots(5.30)$$

which corresponds to the canonical distribution.

Example 5.1 A system with just two energy levels is in thermal equilibrium with a heat reservoir at temperature 600 K. The energy gap between the levels is 0.1 eV. Find (i) the probability that the system is in the higher energy level and (ii) the temperature at which this probability will equal 0.25.

$$(i) \quad P = \frac{e^{-\beta E}}{Z} = \frac{e^{-\beta E}}{1 + e^{-\beta E}} = \frac{1}{e^{\beta E} + 1} = \frac{1}{\frac{1.6 \times 10^{-13}}{e^{1.38 \times 10^{-16} \times 600} + 1}} = 0.126$$

$$(ii) \quad \frac{1}{\frac{1.6 \times 10^{-13}}{e^{1.38 \times 10^{-16} \times T} + 1}} = 0.25$$

$$\text{Therefore} \quad T = 1056 \text{ K}$$

Example 5.2 A solid containing non-interacting paramagnetic atoms, each having a magnetic moment equal to one Bohr magneton, is placed in a magnetic induction field of strength 3 Tesla. Assuming that the atoms are in thermal equilibrium with the lattice, find the temperature to which the solid must be cooled, so that more than 60% of the atoms are polarized with their magnetic moments parallel to the magnetic field.

(One Bohr magneton $\mu = 9.271 \times 10^{-24} \text{ JT}^{-1}$)

Here the probability that the molecules are polarized parallel to the field

$$= \frac{e^{\mu B / kT}}{e^{-\mu B / kT} + e^{\mu B / kT}} = \frac{1}{e^{-2\mu B / kT} + 1} = 0.6$$

$$\text{Therefore} \quad e^{-2\mu B / kT} = \frac{0.4}{0.6}$$

$$\text{i.e.} \quad \frac{2\mu B}{kT} = \ln 1.5$$

$$\text{Therefore} \quad T = \frac{2\mu B}{k \ln 1.5} = 9.91 \text{ K}$$

5.4 MEAN VALUE AND FLUCTUATIONS

We have seen in Ch. 2, that the mean of a function f_α is given by

$$\langle f \rangle = \sum_{\alpha} f_{\alpha} P_{\alpha}$$

In the case of canonical distribution

$$\langle f \rangle = \sum_{\alpha} f_{\alpha} \frac{e^{-\beta E_{\alpha}}}{\sum e^{-\beta E_{\alpha}}} \quad \dots(5.31)$$

Thus, mean energy is given by

$$\langle E \rangle = \sum \frac{E_{\alpha} e^{-\beta E_{\alpha}}}{\sum e^{-\beta E_{\alpha}}} = \sum \frac{E_{\alpha} e^{-\beta E_{\alpha}}}{Z}$$

Thus, $\langle E \rangle = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial}{\partial \beta} (\ln Z) \quad \dots(5.32)$

The mean square energy is

$$\langle E^2 \rangle = \sum \frac{E_{\alpha}^2 e^{-\beta E_{\alpha}}}{\sum e^{-\beta E_{\alpha}}} = \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} \quad \dots(5.33)$$

Therefore, fluctuations in energy is

$$\begin{aligned} \langle (\Delta E)^2 \rangle &= \langle (E - \langle E \rangle)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2 \\ &= \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} - \frac{1}{Z^2} \left(\frac{\partial Z}{\partial \beta} \right)^2 \\ &= \frac{\partial^2}{\partial \beta^2} (\ln Z) = - \frac{\partial \langle E \rangle}{\partial \beta} \end{aligned} \quad \dots(5.34)$$

Using the relation

$$\beta = \frac{1}{kT} \quad \dots(5.35)$$

(to be derived later), we get

$$\langle (\Delta E)^2 \rangle = - \frac{\partial \langle E \rangle}{\partial \beta} = \frac{\partial \langle E \rangle}{\partial T} \frac{\partial T}{\partial \beta} = kT^2 \frac{\partial \langle E \rangle}{\partial T} = kT^2 C_v \quad \dots(5.36)$$

Therefore, the energy fluctuation in the canonical distribution is proportional to the specific heat.

For example, for a perfect gas $E = \frac{3}{2} NkT$

(This will be proved later)

Therefore $\langle (\Delta E)^2 \rangle = kT^2 \frac{3}{2} Nk = \frac{3}{2} Nk^2 T^2 \quad \dots(5.37)$

and the mean fractional fluctuation

$$\left\{ \frac{\langle (\Delta E)^2 \rangle}{\langle E \rangle^2} \right\}^{\frac{1}{2}} = \left\{ \frac{3Nk^2T^2}{2} \frac{4}{9N^2k^2T^2} \right\}^{\frac{1}{2}} = \left(\frac{2}{3N} \right)^{\frac{1}{2}} \quad \dots(5.38)$$

This is negligible if $N \gg 1$.

If $N = 1$, fractional fluctuation is $= \left(\frac{2}{3} \right)^{\frac{1}{2}} = 0.82$, showing that fluctuations are more important for microscopic systems. If $N = 10^{18}$, say, which corresponds to a macroscopic system, fractional fluctuation is 0.82×10^{-9} , which is negligible. It is, therefore, reasonable to assume that in thermal equilibrium a macroscopic system has a precise energy value, which is equal to the ensemble average value of energy.

5.5 GRAND CANONICAL ENSEMBLE

We now consider a situation slightly different from the one considered in the case of canonical distribution. Imagine as before a sub-system A in a system A^1 with a difference that the walls of the sub-systems are such that A and A^1 can exchange both energy and particles. Such systems are known as *open systems*. A^1 is, of course, a member of a microcanonical ensemble. If E_0, N_0 , are the energy and number of particles of the entire system

$$\begin{aligned} E_0 &= E_\alpha + E^1 \\ N_0 &= N_\alpha + N^1 \end{aligned} \quad \dots(5.39)$$

where E_α, E^1 and N_α, N^1 are the energies and number of particles in A and A^1 respectively, and none of them is constant. In deriving the probability distribution, we adopt the same technique as in the case of canonical distribution.

The probability that the sub-system A is in a state with energy E_α and number of particles N_α , is given by

$$P_\alpha = C \Omega_\alpha(E_0 - E_\alpha, N_0 - N_\alpha) \quad \dots(5.40)$$

where C is a constant and Ω_α has the same meaning as in the case of canonical distribution.

We expand $\ln \Omega_\alpha(E_0 - E_\alpha, N_0 - N_\alpha)$ in a power series

$$\begin{aligned} \ln \Omega_\alpha(E_0 - E_\alpha, N_0 - N_\alpha) &= \ln \Omega_\alpha(E_0, N_0) \\ &\quad - \frac{\partial(\ln \Omega_\alpha)}{\partial E_0}, E_\alpha - \frac{\partial(\ln \Omega_\alpha)}{\partial N_0} N_\alpha + \dots \\ &= \ln \Omega_\alpha(E_0, N_0) - \beta E_\alpha + \mu N_\alpha \end{aligned} \quad \dots(5.41)$$

where we have put

$$\beta = \frac{\partial(\ln \Omega_\alpha)}{\partial E_0}, \quad -\mu = \frac{\partial(\ln \Omega_\alpha)}{\partial N_0} \quad \dots(5.42)$$

Hence

$$\Omega_\alpha(E_0 - E_\alpha, N_0 - N_\alpha) = \Omega_\alpha(E_0, N_0) e^{-\beta E_\alpha + \mu N_\alpha}$$

$$\text{Therefore} \quad P_\alpha = C e^{-\beta E_\alpha + \mu N_\alpha} \quad \dots(5.43)$$

The normalization condition gives

$$C = \frac{1}{\sum_\alpha e^{-\beta E_\alpha - \mu N_\alpha \beta}} \quad \dots(5.44)$$

and hence

$$P_\alpha = \frac{e^{-\beta E_\alpha - \mu N_\alpha \beta}}{\sum_\alpha e^{-\beta E_\alpha - \mu N_\alpha \beta}} \quad \dots(5.45)$$

This distribution is called the *grand canonical distribution*.

We now introduce here a quantity

$$z = e^{-\beta \mu} \quad \dots(5.46)$$

which is generally known as the *fugacity* of the system. In the chemical literature it is known as the *absolute activity*.

In terms of fugacity the relation (5.45) becomes

$$P_\alpha = \frac{z^{N_\alpha} e^{-\beta E_\alpha}}{\sum_\alpha z^{N_\alpha} e^{-\beta E_\alpha}} \quad \dots(5.47)$$

The denominator in Eq. (5.47) is called the *grand partition function*.

5.6 ALTERNATIVE DERIVATION OF THE GRAND CANONICAL DISTRIBUTION

We will now show how we can arrive at the distribution given by Eq. (5.45) by following the techniques adopted in Sec. 5.3 for canonical distribution.

As before, the probability that n_1 systems are in state 1 with energy E_1 , n_2 in state 2 with energy E_2 , etc. is given by

$$W = \frac{1}{C} \frac{N!}{n_1! n_2! \dots}$$

where N is the total number of systems in the ensemble. Let N_1 be the number of particles in each of the n_1 systems N_2 in each of the n_2 systems, etc.

Now

$$\ln W = \ln N! - \sum_s \ln n_s! - \ln C$$

$$= N \ln N - N - \sum_s n_s \ln n_s + \sum_s n_s - \ln C$$

The condition for $\ln W$ to be maximum is

$$\delta \ln W = 0$$

$$\text{i.e.} \quad \sum \ln n_s \delta n_s = 0 \quad \dots(5.48)$$

subject to the conditions

$$\sum_s n_s = N; \frac{\sum_s n_s \langle E_s \rangle}{N} = \langle E \rangle; \frac{\sum_s n_s \langle N_s \rangle}{N} = \langle N \rangle$$

$$(i) \sum_s \delta n_s = 0; (ii) \sum_s E_s \delta n_s = 0 \quad (iii) \sum_s N_s \delta n_s = 0.$$

Multiplying (i), (ii) and (iii) by Lagrange's multipliers γ , β and α and adding to Eq. (5.48) we get

$$\sum_s (\ln n_s + \gamma + \beta E_s + \alpha N_s) \delta n_s = 0 \quad \dots(5.49)$$

As δn_s are independent their coefficient must vanish

$$\text{Therefore } \ln n_s + \gamma + \beta E_s + \alpha N_s = 0 \quad \dots(5.50)$$

$$\text{or} \quad n_s = e^{-\gamma - \beta E_s - \alpha N_s} \quad \dots(5.51)$$

$$\text{Now} \quad \sum_s n_s = e^{-\gamma} \sum_s e^{\beta E_s - \alpha N_s} = N$$

$$\text{Therefore} \quad e^{-\gamma} = \frac{N}{\sum_s e^{\beta E_s - \alpha N_s}} \quad \dots(5.52)$$

$$\text{and} \quad n_s = \frac{N e^{-\beta E_s - \alpha N_s}}{\sum_s e^{\beta E_s - \alpha N_s}}$$

The probability of finding the system in state s is given by

$$P_s = \frac{n_s}{N} = \frac{e^{-\beta E_s - \alpha N_s}}{\sum_s e^{\beta E_s - \alpha N_s}} \quad \dots(5.53)$$

which is the same as obtained in the preceding section.

5.7 FLUCTUATIONS IN THE NUMBER OF PARTICLES OF A SYSTEM IN A GRAND CANONICAL ENSEMBLE

The mean square deviation of the number of particles in an open system is given by

$$\langle (\Delta N)^2 \rangle = \langle N^2 \rangle - \langle N \rangle^2 \quad \dots(5.54)$$

$$\text{Now,} \quad \langle N \rangle = \sum_i P_i N_i = \frac{\sum_i N_i e^{-\beta E_i + \beta \mu N_i}}{Z} \quad \dots(5.55)$$

$$\text{and} \quad \langle N^2 \rangle = \frac{1}{Z} \frac{\partial^2 Z}{\partial \alpha^2} \quad \dots(5.56)$$

We also take $\alpha = -\beta\mu$.

$$\text{But} \quad \frac{\partial}{\partial \alpha} \left\{ \frac{1}{Z} \frac{\partial Z}{\partial \alpha} \right\} = \frac{1}{Z} \frac{\partial^2 Z}{\partial \alpha^2} - \frac{1}{Z^2} \left(\frac{\partial Z}{\partial \alpha} \right)^2$$

$$\text{Therefore,} \quad -\frac{\partial \langle N \rangle}{\partial \alpha} = \langle N^2 \rangle - \langle N \rangle^2$$

Hence

$$\langle (\Delta N)^2 \rangle = \frac{\partial \langle N \rangle}{\partial \alpha} \quad \dots(5.57)$$

It will be shown in the next chapter $\alpha = -\beta\mu$, where μ is the chemical potential. Using this relation we can express Eq. (5.57) as

$$\langle (\Delta N)^2 \rangle = kT \frac{\partial \langle N \rangle}{\partial \mu} \quad \dots(5.58)$$

Alternatively, in terms of fugacity z

$$\langle N \rangle = \frac{\sum_i N_i z^{N_i} e^{-\beta E_i}}{Z} = \frac{z}{Z} \frac{\partial Z}{\partial z} = z \frac{\ln Z}{\partial z}$$

$$\begin{aligned} \text{Therefore} \quad z \frac{\partial \langle N \rangle}{\partial z} &= \frac{z}{Z^2} \frac{\partial Z}{\partial z} \sum_i N_i z^{N_i} e^{-\beta E_i} \\ &\quad + \frac{1}{Z} \sum_i N_i^2 z^{N_i} e^{-\beta E_i} \\ &= -\frac{1}{Z^2} \left[\sum_i N_i z^{N_i} e^{-\beta E_i} \right]^2 + \frac{1}{Z} \sum_i N_i^2 z^{N_i} e^{-\beta E_i} \\ &= -\langle N \rangle^2 + \langle N^2 \rangle = \langle (\Delta N)^2 \rangle \quad \dots(5.59) \end{aligned}$$

5.8 REDUCTION OF GIBBS DISTRIBUTION TO MAXWELL AND BOLTZMANN DISTRIBUTIONS

Since we have claimed that Gibbs distribution is a general distribution, we should be able to derive Maxwell and Boltzmann distributions from it. We now proceed to show how this can be done.

5.8.1 Maxwell Distribution

Maxwell distribution function gives basically the probability of finding a particular molecule in a certain range of velocities, say, between u and

$u + du$. The system in this case is a single molecule in a heat reservoir. The Γ space is a six-dimensional space, i.e. the M-space.

Since a molecule occupies in the phase space a volume h^3 , the number of states of a molecule contained in a phase volume element $d\Gamma$ is

$$\frac{d\Gamma}{h^3} = \frac{dx dy dz dp_x dp_y dp_z}{h^6} \quad \dots(5.60)$$

where h is Planck's constant.

According to Gibbs distribution the probability of a system being in a state with energy E_α is $\frac{e^{-\beta E_\alpha}}{\sum e^{-\beta E_\alpha}} g_\alpha$.

Therefore, the probability of finding the system in a phase volume element $d\Gamma$ corresponding to the energy E_α is

$$\begin{aligned} dP_\alpha &= \frac{e^{-\beta E_\alpha}}{\sum e^{-\beta E_\alpha}} \frac{d\Gamma}{h^3} \\ &= \frac{e^{-\beta E_\alpha}}{\sum e^{-\beta E_\alpha}} \frac{dx dy dz dp_x dp_y dp_z}{h^3} \quad \dots(5.61) \end{aligned}$$

To find the probability that the molecule has energy E_α , we have to integrate Eq. (5.61) with respect to all the elements of the phase volume corresponding to E_α . Since the energy does not depend upon the coordinates, we can straight away integrate Eq. (5.61) and write

$$dP_\alpha = \frac{V}{h^3} \frac{e^{-\beta E_\alpha}}{\sum e^{-\beta E_\alpha}} dp_x dp_y dp_z \quad \dots(5.62)$$

By converting into spherical coordinates and integrating over all the angular coordinates we have

$$dP_\alpha = \frac{4\pi V}{h^3} \frac{e^{-\beta E_\alpha}}{\sum e^{-\beta E_\alpha}} p^2 dp \quad \dots(5.63)$$

Now $E = \frac{1}{2} mu^2; p = mu$

Substituting these in Eq. (5.63), we have

$$\begin{aligned} dP_\alpha &= \frac{4\pi V}{h^3} \frac{e^{-\beta mu^2/2}}{\sum e^{-\beta mu^2/2}} m^2 u^2 m du \\ &= \frac{4\pi V m^3}{h^3} \frac{e^{-\beta mu^2/2}}{\sum e^{-\beta mu^2/2}} u^2 du \quad \dots(5.64) \end{aligned}$$

The normalization condition gives $\int_0^\infty dP_\alpha = 1$.

That is

$$\frac{4\pi Vm^3}{h^3 \sum e^{-\beta mu^2/2}} \int_0^\infty e^{-\beta mu^2/2} u^2 du = 1$$

i.e.

$$\begin{aligned} & \frac{4\pi Vm^3}{h^3 \sum e^{-\beta mu^2/2}} \frac{1}{2\left(\frac{\beta m}{2}\right)^{\frac{3}{2}}} \Gamma\left(\frac{1}{2}\right) \\ &= \frac{4\pi Vm^3}{h^3 \sum e^{-\beta mu^2/2}} \frac{\sqrt{2}}{(\beta m)^{\frac{3}{2}}} \frac{\sqrt{\pi}}{2} \\ &= \frac{4\pi Vm^3}{h^3 \sum e^{-\beta mu^2/2}} \left(\frac{2}{\beta m}\right)^{\frac{3}{2}} \frac{\sqrt{\pi}}{4} = 1 \end{aligned}$$

Therefore

$$= \frac{4\pi Vm^3}{h^3 \sum e^{-\beta mu^2/2}} = \frac{4}{\sqrt{\pi}} \left(\frac{\beta m}{\sqrt{\pi}}\right)^{\frac{3}{2}} \quad \dots(5.65)$$

Hence

$$dp_\alpha = \frac{4}{\sqrt{\pi}} \left(\frac{\beta m}{2}\right)^{\frac{3}{2}} e^{-\beta mu^2/2} u^2 du \quad \dots(5.66)$$

This is Maxwell distribution. Comparing this with Eq. (3.23), we see that the two relations differ only in the constants appearing in them and will be identical if

$$\beta = 2\alpha m = \frac{2m}{2mkT} = \frac{1}{kT} \quad \dots(5.67)$$

This relation can also be proved by using Eq. (5.67) to calculate mean kinetic energy, as

$$\begin{aligned} \left\langle \frac{mu^2}{2} \right\rangle &= \int_0^\infty \frac{1}{2} mu^2 \frac{4}{\sqrt{\pi}} \left(\frac{\beta m}{2}\right)^{\frac{3}{2}} e^{-\beta mu^2/2} u^2 du \\ &= \frac{2m}{\sqrt{\pi}} \left(\frac{\beta m}{2}\right)^{\frac{3}{2}} \int_0^\infty e^{-\beta mu^2/2} u^4 du \\ &= \frac{2m}{\sqrt{\pi}} \left(\frac{\beta m}{2}\right)^{\frac{3}{2}} \frac{1}{2} \left(\frac{2}{\beta m}\right)^{\frac{5}{2}} \frac{3}{4} \sqrt{\pi} = \frac{3}{2\beta} \end{aligned} \quad \dots(5.68)$$

Using the result proved later

$$\left\langle \frac{1}{2} mu^2 \right\rangle = \frac{3}{2} kT = \frac{3}{2\beta} \quad \dots(5.69)$$

We have

$$\beta = \frac{1}{kT} \quad \dots(5.70)$$

Hence, Maxwell distribution is

$$dP(u) = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2kT} \right)^{\frac{3}{2}} e^{-\left(\frac{m}{2kT} \right) u^2} u^2 du \quad \dots(5.71)$$

which is the same as Eq. (3.23).

5.8.2 Boltzmann Distribution

If the gas is an external potential field, the particle energy E_α is given by

$$E_\alpha = \frac{1}{2} m u^2 + U(x, y, z) \quad \dots(5.72)$$

where $U(x, y, z)$ is potential energy of the external field. Therefore

$$dP(x, y, z, p_x, p_y, p_z) = \frac{e^{-\beta \left\{ \frac{1}{2} m u^2 + U \right\}}}{\sum e^{-\beta \left\{ \frac{1}{2} m u^2 + U \right\}}} \frac{dx dy dz dp_x dp_y dp_z}{h^3} \quad \dots(5.73)$$

Since the probability densities of coordinates and momenta of a particle are independent

$$dP(x, y, z, p_x, p_y, p_z) = dP_1(x, y, z) dP_2(p_x, p_y, p_z)$$

where

$$dP_1(x, y, z) = C_1 e^{-\beta U(x, y, z)} dx dy dz \quad \dots(5.74)$$

and

$$dP_2(p_x, p_y, p_z) = C_2 e^{-\beta m u^2 / 2} dp_x dp_y dp_z \quad \dots(5.75)$$

Here C_1, C_2 are the constants to be found from the normalization condition. The second of these equations was discussed in the preceding section and leads to Maxwell distribution. We will now show that the first leads to Boltzmann distribution.

Equation (5.74) gives the probability that a molecule is located in the volume element $dx dy dz$ at the point (x, y, z) . If N is the number of molecules in the system, the number of molecules in the volume element $dx dy dz$ is

$$dn(x, y, z) = C_1 N e^{-\beta U(x, y, z)} dx dy dz \quad \dots(5.76)$$

Therefore, the concentration at the point (x, y, z) is

$$n_0(x, y, z) = \frac{dn}{dx dy dz} = C_1 N e^{-\beta U} \quad \dots(5.77)$$

Suppose, at a given point (x_0, y_0, z_0) the concentration is $n_0(x_0, y_0, z_0)$ and potential energy is $U(x_0, y_0, z_0)$. Therefore

$$n_0(x_0, y_0, z_0) = C_1 N e^{-\beta U(x_0, y_0, z_0)} \quad \dots(5.78)$$

From Eqs. (5.77) and (5.78) we get

$$\frac{n_0(x, y, z)}{n_0(x_0, y_0, z_0)} = e^{-\beta [U(x, y, z) - (x_0, y_0, z_0)]} \quad \dots(5.79)$$

$$\text{i.e.} \quad n_0(x, y, z) = n_0(x_0, y_0, z_0) e^{-\Delta U/kT} \quad \dots(5.80)$$

This is Boltzmann distribution.

Example 5.3 Calculate the average potential energy of a molecule of an ideal gas in thermal equilibrium at absolute temperature T , contained in a cubical box of side L , the only external field acting on the gas being the earth's uniform gravitational field.

Let m be the mass of a molecule. Hence, its potential energy at height z from the bottom of the box is $U = mgz$. Therefore

$$\begin{aligned} \langle U \rangle &= \frac{\int_0^L mgz e^{-\beta mgz} dz}{\int_0^L e^{-\beta mgz} dz} \\ \text{Numerator} &= mg \left[\frac{ze^{-\beta mz}}{-mg\beta} - \frac{e^{-\beta mgz}}{\beta^2 m^2 g^2} \right]_0^L \\ &= -\frac{Le^{-\beta mgL}}{\beta} + \frac{1le^{-\beta mgL}}{\beta^2 mg} \\ \text{Denominator} &= \frac{1 - e^{-\beta mgL}}{\beta mg} \\ \langle U \rangle &= -\frac{mgLe^{-\beta mgL}}{1 - e^{-\beta mgL}} + \frac{1}{\beta} \\ &= \frac{mgL}{1 - e^{-\beta mgL}} + kT \end{aligned}$$

5.9 BAROMETRIC FORMULA

As an application of the Boltzmann distribution, we shall now show how it can be used to obtain a formula describing the variation of the atmospheric pressure with altitude. We assume that the atmosphere is in equilibrium and its temperature is constant, which, incidentally, is only an approximate assumption.

If m is the mass of a molecule and h is the height at which the concentration is to be found, the potential energy of the molecule at this height

$$\Delta U = mgh \quad \dots(5.81)$$

The concentration at this altitude is

$$n(h) = n(0) e^{-\beta \Delta U} = n(0) e^{-mgh/kT} \quad \dots(5.82)$$

At constant temperature, the pressure is proportional to concentration. Hence, we can write Eq. (5.82) in the form

$$p(h) = p(0) e^{-mgh/kT} \quad \dots(5.83)$$

where $p(h)$ is the pressure at altitude h and $p(0)$ the pressure at $h = 0$.

This is known as *Barometric formula*.

5.10 EXPERIMENTAL VERIFICATION OF THE BOLTZMANN DISTRIBUTION

Take a vessel with a liquid filled with a powder of very fine particles, whose density is slightly higher than that of the liquid. The particles under the force of gravity will be distributed in height non-uniformly in accordance with Eq. (5.82). The ratio of the number of particles at height h and at the bottom is

$$\frac{n(h)}{n(0)} = e^{-mgh/kT} \quad \dots(5.84)$$

where mg is the weight of the particle in the liquid. If ρ is the density of the particle, r its radius and ρ_0 the density of the liquid

$$mg = \frac{4}{3} \pi r^3 (\rho - \rho_0) g \quad \dots(5.85)$$

where account is taken of the buoyant force acting on the particle.

$$\text{Therefore} \quad \frac{n(h)}{n(0)} = e^{-\frac{4}{3} \pi r^3 (\rho - \rho_0) gh / kT} \quad \dots(5.86)$$

The number of particles at various heights can be measured with the help of a microscope. One can verify whether the concentrations thus measured satisfy Boltzmann distribution.

Experiments of this type were carried out by J Perrin, the results of which confirmed Boltzmann distribution.

The relation (5.86) can be expressed as

$$k = \frac{4/3 \pi r^3 (\rho - \rho_0) gh}{T \ln \frac{n(0)}{n(h)}} \quad \dots(5.87)$$

Perrin used this relation to obtain the value of k .

5.11 MIXTURE OF GASES

We have derived Eqs. (5.82) and (5.83) for the variation of concentration and pressure for a single type of particles. We often come across a situation in which two or more types of particles are present. For example, Earth's atmosphere is mainly a mixture of two gases: nitrogen and oxygen. We shall discuss in this section the variation of concentration in the mixture of two gases.

Suppose a cylindrical vessel with base area A and height h , contains two types of molecules. Let their total number be N_1 and N_2 and mass of a molecule of each type be m_1 and m_2 respectively. Since the probability that a molecule is located at a particular point is independent of the positions of all other molecules, we can take the distribution of each type

of molecules as

$$\begin{aligned}n_1(h) &= A_1 n_1(0) e^{-m_1 g h / kT} \\n_2(h) &= A_2 n_2(0) e^{-m_2 g h / kT}\end{aligned}\quad \dots(5.88)$$

where $n_1(h)$, are the concentration at height h and A_1 are constants.

From the normalization condition

$$A_1 \int_0^{h_0} n_1(0) e^{m_1 g h / kT} dh = N_1$$

$$\text{i.e.} \quad A_1 n_1(0) \left[-\frac{kT}{m_1 g} e^{m_1 g h / kT} \right]_0^{h_0} = N_1$$

$$\text{or,} \quad A_1 n_1(0) \frac{kT}{m_1 g} \{1 - e^{m_1 g h_0 / kT}\} = N_1$$

$$\text{Therefore} \quad A_1 = \frac{N_1 m_1 g}{n_1(0) kT \{1 - e^{-m_1 g h_0 / kT}\}} \quad \dots(5.89)$$

$$\text{and} \quad n_1(h) = \frac{N_1 m_1 g}{kT} \frac{e^{-m_1 g h / kT}}{\{1 - e^{-m_1 g h_0 / kT}\}} \quad \dots(5.90)$$

$$\text{Similarly} \quad n_2(h) = \frac{N_2 m_2 g}{kT} \frac{e^{-m_2 g h / kT}}{\{1 - e^{-m_2 g h_0 / kT}\}} \quad \dots(5.91)$$

These formulae show that the concentration of the heavier molecules decreases rapidly with height.

The ratio of the concentration is

$$\frac{n_2(h)}{n_1(h)} = \frac{N_2 m_2}{N_1 m_1} \frac{\{1 - e^{-m_1 g h_0 / kT}\}}{\{1 - e^{-m_2 g h_0 / kT}\}} \quad \dots(5.92)$$

Example 5.4 The Earth's atmosphere at sea level consists of 78% nitrogen and 22% oxygen by volume (The presence of other components is neglected.) Find the percentage content of these gases at an altitude of 8.75 km, assuming the temperature of the atmosphere to be 300K.

The concentrations of the two gases are

$$n_{O_2}(h) = n_{O_2}(0) e^{-m_{O_2} g h / kT}$$

$$n_{N_2}(h) = n_{N_2}(0) e^{-m_{N_2} g h / kT}$$

If N is the concentration of the mixture at the sea level

$$n_{O_2}(0) = 0.22N; \quad n_{N_2}(0) = 0.78N$$

The percentage of oxygen content at h is

$$\begin{aligned}
 &= 100 \frac{0.22 Ne^{-m_{O_2}gh/kT}}{0.22 Ne^{-m_{O_2}gh/kT} + 0.78 Ne^{-m_{N_2}gh/kT}} \\
 &= \frac{0.22}{0.22 + 0.78e^{-(m_{N_2}-m_{O_2})gh/kT}} \\
 &= 22 \left\{ 1 - \frac{0.78(m_{N_2} - m_{O_2})gh}{kT} \right\}
 \end{aligned}$$

$$m_{O_2} = 5344 \times 10^{-27} \text{ kg}; \quad m_{N_2} = 47.76 \times 10^{-27} \text{ kg}$$

Substituting these values, we get oxygen content in per cent as 19.8%. Hence nitrogen content is 80.2%.

While closing this chapter we have to draw the attention of the reader to the fact that the parameter α has not yet been identified. This will be done in the next chapter.

PROBLEMS

- 5.1 Evaluate the partition function at temperature T for a classical one-dimensional oscillator. Hence find the mean energy of the oscillator.
- 5.2 A small system has just two states of energy $E_1 = 0$ and $E_2 = 10^{-22}$ J. Find the probabilities p_1 and p_2 for the system to be in states of 1 and 2 respectively, when the mean energy $\langle E \rangle$ of the system is (i) $0.2 E_2$, (ii) $0.5 E_2$. Assuming Boltzmann distribution calculate the temperature in the two cases.
- 5.3 A rectangular box of maximum volume is to be made out of a cardboard of fixed area A . Using the method of Lagrange multipliers, show that the box with maximum volume is a cube.
- 5.4 Show that the specific heat of an ideal gas in the extreme relativistic case is twice its value for unrelativistic monatomic gas.
- 5.5 The number density of magnetic ions in a paramagnetic salt is $10^{24}/\text{m}^3$. Each ion has a magnetic moment of one Bohr magneton, which can always be parallel or anti-parallel to the external field. Calculate (i) the difference between the number of ions in these two states, (ii) the mean magnetic energy, if the magnetic field applied is 1 T and the temperature 3 K.
- 5.6 An infinite column of a classical gas consisting of N identical atoms each of mass m is at thermal equilibrium in a uniform gravitational field. Find the mean energy per atom.
- 5.7 A cubical box of side L contains an ideal gas of N molecules, each of mass m , in thermal equilibrium at absolute temperature T . The only external influence acting on the system is the Earth's

uniform gravitational field. Calculate the average kinetic energy of a molecule.

- 5.8 Assuming that the atmosphere is all at one temperature T , find the mean height of a gas molecule in the atmosphere.
- 5.9 Find the height at which the atmospheric pressure is $\frac{1}{100}$ th that of the sea level. Assume that the atmosphere is at a constant temperature 300 K.
- 5.10 Gamboge gum particles were added to a vessel containing water. The density of gum particles is $1.21 \times 10^3 \text{ kg/m}^3$ and the volume of each grain is $1.03 \times 10^{-19} \text{ m}^3$. Find the height at which the concentration of grain distribution decreases to half its value at the bottom of the vessel, the measurements being carried out at 4°C .
- 5.11 Following data on the number of Gamboge particles each of mass $9.8 \times 10^{-16} \text{ kg}$ and density 1350 kg/m^3 in water, was obtained in an experiment on gravitational sedimentation equilibrium:
- | | | | | | |
|--------------------------|-----|-----|-----|-----|-----|
| Height in μ | 0 | 25 | 50 | 75 | 100 |
| Mean number of particles | 200 | 175 | 150 | 130 | 110 |
- Calculate the temperature of water.

